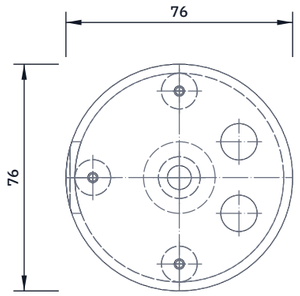
BUILD PLAN SKETCH, PLAN NOT YET BUILT

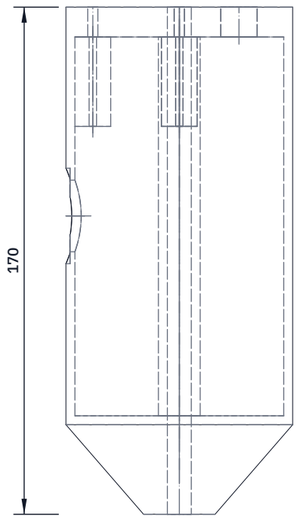
GrainGuard · github.com/BoujeeEnjinia1701/grainguard

REV	DESCRIPTION	DATE	BY
P1	Making sketch for the prototype build plan	2026-10-01	AC



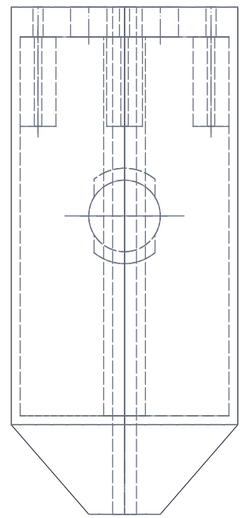
TOP VIEW

Scale 1:2; looking down (along -Z)



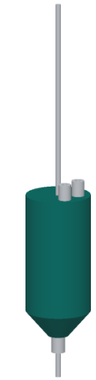
FRONT VIEW

Scale 1:2; looking at the front (along +Y)



RIGHT VIEW

Scale 1:2; looking at the right side (along -X)



WHERE IT GOES

This part in colour, its neighbours in grey

HOW TO MAKE IT AND HOW IT FITS

Body: 76 mm across, 130 mm from the rim to the start of the cone,
a 30 mm cone below to a 24 mm tip; 3 mm walls and floor.
A 14 mm tube runs up the middle from tip to rim, bore 8 mm for the rope.
Side window 24 mm across, 60 mm below the rim, opposite the glands,
with a 32 mm recess 1.4 mm deep round it for the PTFE membrane.
Three screw bosses 12 mm across, 29 mm from the centre at 90, 180 and
270 degrees, 30 mm deep from the rim, 2.5 mm holes for M3 screws.
Lid: 76 mm across, 10 mm thick, 8 mm centre hole; two 12.2 mm holes
20 mm out and 12 mm each side of the centre line for M12 glands;
three 3.4 mm holes over the bosses. 1 mm EPDM gasket under the lid.
Print the body upright, cone down; print the lid flat. ASA, not PLA.
Fit: the pod slides on the rope and sits on a rope stop under its tip.
Check: the rope slides through freely; the lid seats flat on the gasket.

GRAINGUARD

github.com/BoujeeEnjinia1701/grainguard

GrainGuard sensor pod body and lid (make 6 of each): making sketch

DRAWING NO.	REV	SHEET	SIZE
GGD-DWG-101	P1	1 of 1	ANSI B

SCALE	UNITS	PROJECTION	DATE	DRAWN
1:2	mm		2026-10-01	Amish Chadha

MATERIAL / NOTES

ASA, 3D printed, 4 walls, 40 % infill

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